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Q What is autotrophs & heterotrophs?

Ans: • Autotrophs are organisms that can produce their own food from inorganic sources using light or chemical energy. They are the producers in an ecosystem. A classic example is a plant using sunlight for photosynthesis.

• Heterotrophs are organisms that cannot produce their own food and must obtain energy and nutrients by consuming other organisms. They are the consumers or decomposers. Animal, fungi, and most bacteria are examples.

Q Define Taxonomy?

Ans:- Taxonomy is the scientific discipline of naming, defining and classifying groups of biological organisms based on shared characteristics. It creates a hierarchical system that helps us understand the evolutionary relationships between different life forms.

Q What is cell?

Ans: A cell is the most basic structural, functional and biological unit of all known living organisms. Often called the "building block of life" it is the smallest unit of life that can replicate independently.

Q What are the three domains of life?

Ans:- The three domains, which are the highest level of biological classification, are:

(i) Archaea → These are single celled prokaryotic microorganisms. Many archaea are extremophiles, meaning they live in extreme environments like hot springs or salt flats.

(ii) Bacteria → This domain consists of single-celled prokaryotic microorganisms that are found almost everywhere on Earth, including in and on our bodies.

(iii) Eukarya → This domain includes all organisms whose cells contain a true nucleus and other membrane-bound organelles. It encompasses four kingdoms: Protista, Fungi, Plantae (plants), and Animalia (animals)

Q Define photosynthesis with the help of chemical reaction.

Ans: Photosynthesis is the process used by plants, algae and some bacteria to convert light energy into chemical energy, which is stored in the bonds of sugar molecules. This process uses carbon dioxide and water and releases oxygen as a byproduct.

The overall balanced chemical equation for photosynthesis is:

$$6CO_2 + 6H_2O \xrightarrow{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

Q Explain about hierarchy classification of living organism.

Ans: Biological classification organizes living things into a hierarchical system of groups, with each level representing a taxon of increasing specificity. The main levels, from broadest to most specific: This system helps scientists understand the relationship between different organisms and how they have evolved: Here's a breakdown of the levels.

- Domain: The highest level of classification, grouping organisms based on fundamental cell types.
- Kingdom: Divides living things into broad categories like Animalia, Plantae, Fungi, Protista, and Monera
- Phylum (or Division): Groups organisms with similar body plans, For example, all animals with a backbone are in the phylum Chordata.
- Class: Further divides organisms with a ~~similar bod~~ phylum based on more specific characteristics.
- Order: A group of similar families
- Family: A group of similar genera
- Genus: A group of closely related species.
- Species: The most specific level, defining a group of organisms that can interbreed and produce fertile offspring.

Domain (Domains)
Kingdom (Kingdoms)
Phylum (Phyla)
Class (Classes)
Order (Orders)
Family (Families)
Genus (General
Species (Species)

Q How autotrophs utilize carbon and energy?

Ans: Autotrophs build their own organic molecules from inorganic sources. They do this by combining a source of energy with a source of carbon.

- Energy Source:

* Photoautotrophs: Use light energy from the sun (e.g., plants algae)
* Chemoautotrophs: Use chemical energy from the oxidation of inorganic compounds like hydrogen sulfide or ammonia (e.g., certain bacteria in deep-sea vents).

- Carbon Source:- All autotrophs use inorganic carbon, primarily carbon dioxide from atmosphere or water. They "fix" this CO_2 into organic compounds like glucose through processes like photosynthesis or chemosynthesis. This stored chemical energy, then fuels their metabolic activities.

Q Compare the characteristics of Prokaryotic and Eukaryotic cell.

Ans

Feature	Prokaryotic Cell	Eukaryotic Cell
Nucleus	Absent; genetic material is in a region called the nucleoid.	Present; a true nucleus enclosed by a membrane.
Membrane-Bound Organelles	Absent (e.g., no mitochondria, ER, Golgi).	Present (e.g., mitochondria, ER, Golgi, lysosomes)
DNA Structure	Single, circular chromosome	Multiple, linear chromosomes.
Size	Generally much smaller (0.1 - 5.0 μm).	Generally much larger (10 - 100 μm).
Cell Division	Binary Fission (simpler)	Mitosis and Meiosis (more complex)
Ribosomes	Smaller (70S).	Larger (80S).
Examples	Bacteria, Archaea.	Plants, Animal, Fungi, Protists.

Q Explain mode of excretion in Uricotelic organisms.

Ans Uricotelic organisms are those that ~~exercise~~ excrete excess nitrogen primarily as uric acid.

- Process - Uric acid is complex, non-toxic, and largely insoluble in water. It is excreted as a semi-solid paste or a dry powder, which requires very little water for its removal.

- Advantage - This mode of excretion is a significant adaption for water conservation making it ideal for animals living in arid environments or for those, like birds and insects, where minimizing weight is crucial.

Examples:- Birds, reptiles, insects and land snails.

Q Explain the classification of organisms based on carbon utilization of organisms.

Ans Organisms are broadly classified into two major groups based on how they obtain the carbon needed to build their cellular structure.

(i) Autotrophs → These are "self-feeders" that use inorganic carbon (mainly carbon dioxide, CO_2) as their carbon source. They convert this inorganic carbon into organic compounds.

(ii) Heterotrophs → These are "other-feeders" that must obtain carbon from organic compounds by consuming other organisms or their by-products. Humans are heterotrophs, getting carbon from the food we eat.

These categories can be further combined with how they obtain energy to form four main metabolic groups: Photoautotrophs, Chemoautotrophs, Photoheterotrophs and Chemoheterotrophs.

Q Differentiate between Unicellular and Multicellular organisms.

Feature	Unicellular Organisms	Multicellular Organisms
Number of cells	Consist of a single cell	Consist of many cells.
Body organization	The single cell performs all life functions (feeding, respiration, excretion, reproduction).	Cells are specialized and organized into tissues, organs, and organ systems.
Division of Labor	No division of labor, one cell does everything	High degree of labor division among specialized cells.
Lifespan	Typically very short	Generally much longer.
Injury	Injury to the cell usually leads to the death of the organism	The organism can survive the death of some cells and can often repair or replace them.
Examples:-	Bacteria, Amoeba, Yeast, Paramecium	Humans, Trees, insects, Fungi.